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MIDTERM TEST

ECE 657A - Section 2 - Data and Knowledge Modelling and Analysis

University of Waterloo

Term: Winter Year: 2022

Date: Tuesday, March 1, 2022

Time: 10:00 – 11:20 pm (80 minutes)

Instructors: Prof. Mark Crowley

Lecture Section: 002

Exam Type: In person written test

Number of exam pages (includes cover page): 16

Additional Materials Allowed: Notes (3 pages, handwritten, double-sided) and calculator allowed

Please Print

UWaterloo Quest ID _____

Last Name _____

First Name _____

Username _____

Section _____

Instructions

1. Answer the questions in the spaces provided. If you require additional space to answer a question, the last two pages and refer to those pages in your solutions. Do not tear off the pages in the booklet.
2. Use a dark pencil or pen for your work.
3. For multiple choice and true-false questions (A=True, B=False) fill in the right answer on the bubble sheet on the final page.
4. During the text, questions will only be answered for clarification of meaning or if you think there is a mistake. For anything more, or if the question is still unclear then state your assumptions and understanding of the question on the page and answer as best you can.
5. Washroom breaks: this is a short test, so hopefully it will not be necessary to be excused for a washroom break. However, if you need to, raise your hand and a TA will escort you to the outside of the washroom, leave your phone in your bag.
6. Additional Materials Allowed: Notes (3 pages, handwritten, double-sided) and calculator for page numbers.

True or False

Answer the following questions with either True or False. Enter your answer on the **bubble sheet** at the back of your exam, use "A" for True and use "B" for False. Note: you can circle the correct answer on the question sheet as well for reference, but be sure to fill in the correct option on the bubble sheet.

1. **True ("A") / False ("B") :** If your dataset is **large**, you would be better off with a model that has **low variance** and **high bias** rather than one with high variance and low bias.
2. **True ("A") / False ("B") :** If your dataset is **small**, you would be better off with a model that has **low variance** and **high bias** rather than one with high variance and low bias.
3. **True ("A") / False ("B") :** Computing the MLE for linear regression can yield multiple local optima.
4. **True ("A") / False ("B") :** A MAP estimator is equivalent to the MLE estimator when the prior used in the MAP is a uniform prior over the parameter space.
5. **True ("A") / False ("B") :** AdaBoost will continue improving indefinitely eventually give zero training error regardless of the type of weak classifier it uses, provided enough iterations are performed.

Multiple Choice

Choose the correct answer to the following questions. Enter your answer on the **bubble sheet** at the back of your exam for the corresponding question number and letter. Note: you can circle the correct answer on the question sheet as well for reference, but be sure to fill in the correct option on the bubble sheet.

6. Which is of the following situations gives you better confidence your model is learning well?
 - (A) A classifier that achieves 94% accuracy on the training set and 70% accuracy on test set
 - (B) A classifier that achieves 70% accuracy on the training set and 75% accuracy on test set.
7. What concept does the previous demonstrate about your model? (write the answer on this sheet)
8. When data is divided into training, validation and testing sets, and changing some parameter is leading to improved accuracy on the training set, but decreasing accuracy on the validation set, then what is the explanation?
 - (A) there is not enough data for training
 - (B) the model is biased in some way
 - (C) the data probably has too many outliers
 - (D) the model is overfitting to the training data
 - (E) the model is underfitting to the training data
9. MAP estimators are less likely to suffer from over-fitting compared to MLE estimators due to which of the following?
 - (A) the use of Bayes rule
the prior distribution
 - (B) the use of the expectation operator
 - (C) no assumption about features being independent
 - (D) the posterior distribution

Multiple Choice Data Preparation

We learned about a number of data pre-processing methods in class, such as the following:

- (A) Cross-validation
- (B) Histograms
- (C) Normalization
- (D) Interpolation
- (E) One-hot Encoding

For each of the numbered tasks or situations below, choose the letter corresponding to the most appropriate data pre-processing method to use:

- 10. Using both Nominal and Continuous Data

(A)	(B)	(C)	(D)	(E)
-----	-----	-----	-----	-----
- 11. Noisy Data

(A)	(B)	(C)	(D)	(E)
-----	-----	-----	-----	-----
- 12. Missing Data

(A)	(B)	(C)	(D)	(E)
-----	-----	-----	-----	-----
- 13. Finding Outliers

(A)	(B)	(C)	(D)	(E)
-----	-----	-----	-----	-----
- 14. Different Data Scales

(A)	(B)	(C)	(D)	(E)
-----	-----	-----	-----	-----
- 15. Reducing Bias of Your Model

(A)	(B)	(C)	(D)	(E)
-----	-----	-----	-----	-----

Multiple Choice Algorithm Assessment

You are asked to prepare an assessment of the available methods to use for the following problems. For each numbered problem, there are a number of choices indicated by letter in the table. Check off the appropriate selection for each. Then go to the bubble sheet at the end of the test booklet and fill in the appropriate lettered boxes for each numbered question.

The table options are as follows, select one case only for each algorithm. Provide a one line explanation for cases (B) and (C) only:

- (A) This method could be used, without additional assumptions.
- (B) This method could be used, with some additional assumptions (explain).
- (C) Not applicable to this task for specific reasons (explain).
- (D) This method doesn't do the desired task at all

II: Sensor data on energy, heat, noise and speed for a factory assembly line motor, where the general relationship between the variables is known. The task is to monitor the outcoming sensor data and learn a model that can predict one of the variables from the other three.

16. MLE

(A)	(B)	(C)	(D)
-----	-----	-----	-----

17. Explanation (if needed):

18. MAP

(A)	(B)	(C)	(D)
-----	-----	-----	-----

19. Explanation (if needed):

20. EM

(A)	(B)	(C)	(D)
-----	-----	-----	-----

21. Explanation (if needed):

22. Naive Bayes

(A)	(B)	(C)	(D)
-----	-----	-----	-----

23. Explanation (if needed):

24. PCA

(A)	(B)	(C)	(D)
-----	-----	-----	-----

25. Explanation (if needed):

26. LDA

(A)	(B)	(C)	(D)
-----	-----	-----	-----

27. Explanation (if needed):

28. kNN

(A)	(B)	(C)	(D)
-----	-----	-----	-----

29. Explanation (if needed):

30. t-SNE

(A)	(B)	(C)	(D)
-----	-----	-----	-----

31. Explanation (if needed):

Pre-processing with PCA

Consider the use of PCA as a pre-processing step on your data for a classification task and answer the following questions briefly. Let our dataset be a moderate size dataset $\mathbf{X} \in \mathbb{R}^d$ with 50 features, which are very human interpretable, a numerical label that relates in a complex, non-linear way to the features, and 10,000 data points. We apply PCA to the data obtain the eigenvector matrix $\mathbf{U} \in \mathbb{R}^{d \times d}$ without truncation.

We then use the projection $\mathbf{U}^\top \mathbf{X}$ to train each of the following models for our classification problem, we compare the accuracy of the classifier against being trained on $\mathbf{X}$ directly.

32. Naive Bayes: What is one benefit and one drawback of using this PCA projection in this case?

33. Decision Tree using C4.5: What is one benefit and one drawback of using this PCA projection in this case?

34. Random Forests with 100 tree: What is one benefit and one drawback of using this PCA projection in this case?

Confusion Matrix Calculations

Calculate the answer and enter in the answer box with at least two decimal points of precision.

Confusion Matrix

Imagine you have built a classifier for identifying cars, and it correctly identifies 41 cars and 56 trucks. However, it incorrectly labels 8 cars as trucks, and it incorrectly labels 12 trucks as cars.

35. Draw the confusion matrix for this situation where the "positive" class is cars.

Accuracy

36. What is the accuracy of the classifier? Show your calculation and the answer.

Recall

37. What is the the recall for the classifier? Show your calculation and the answer.

Representation Learning

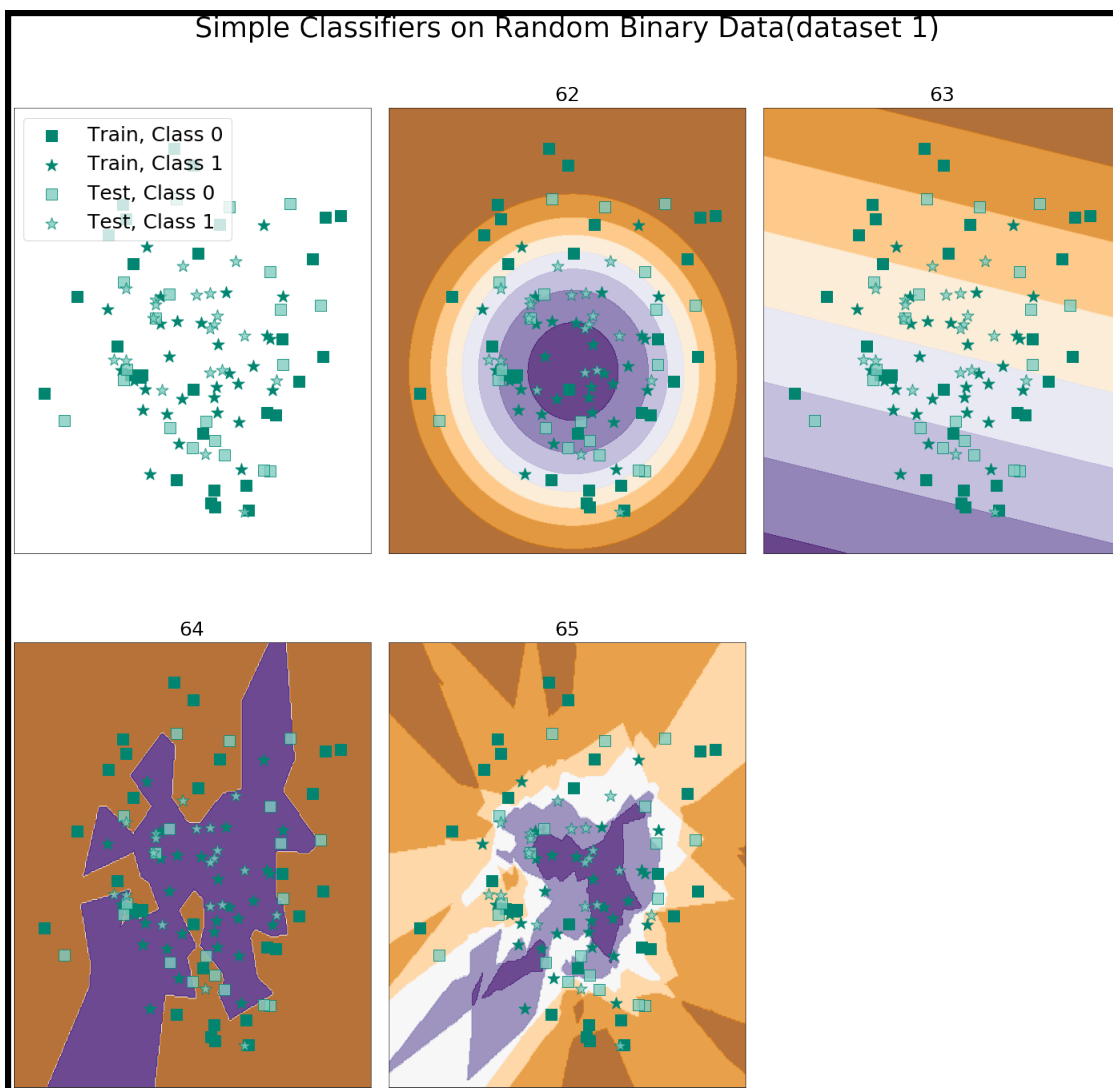
38. Why is Singular Value Decomposition (SVD) an important alternative way to perform Principle Component Analysis (PCA)?
- (A) SVD is an approximation of the PCA solution, so it is more generalizable.
 - (B) SVD doesn't require the data matrix to be centred.
 - (C) SVD can deal with missing data.
 - (D) SVD is more computationally efficient to compute than PCA in some circumstances.

FDA vs. LDA

39. What is the difference between FDA and LDA at a high level?
- (A) FDA is supervised and LDA is not.
 - (B) LDA uses the features extracted by FDA to define a classifier.
 - (C) LDA focusses on classification whereas FDA focusses on separation and variance.
 - (D) LDA is computationally more efficient than FDA.

Algorithm Decision Boundaries

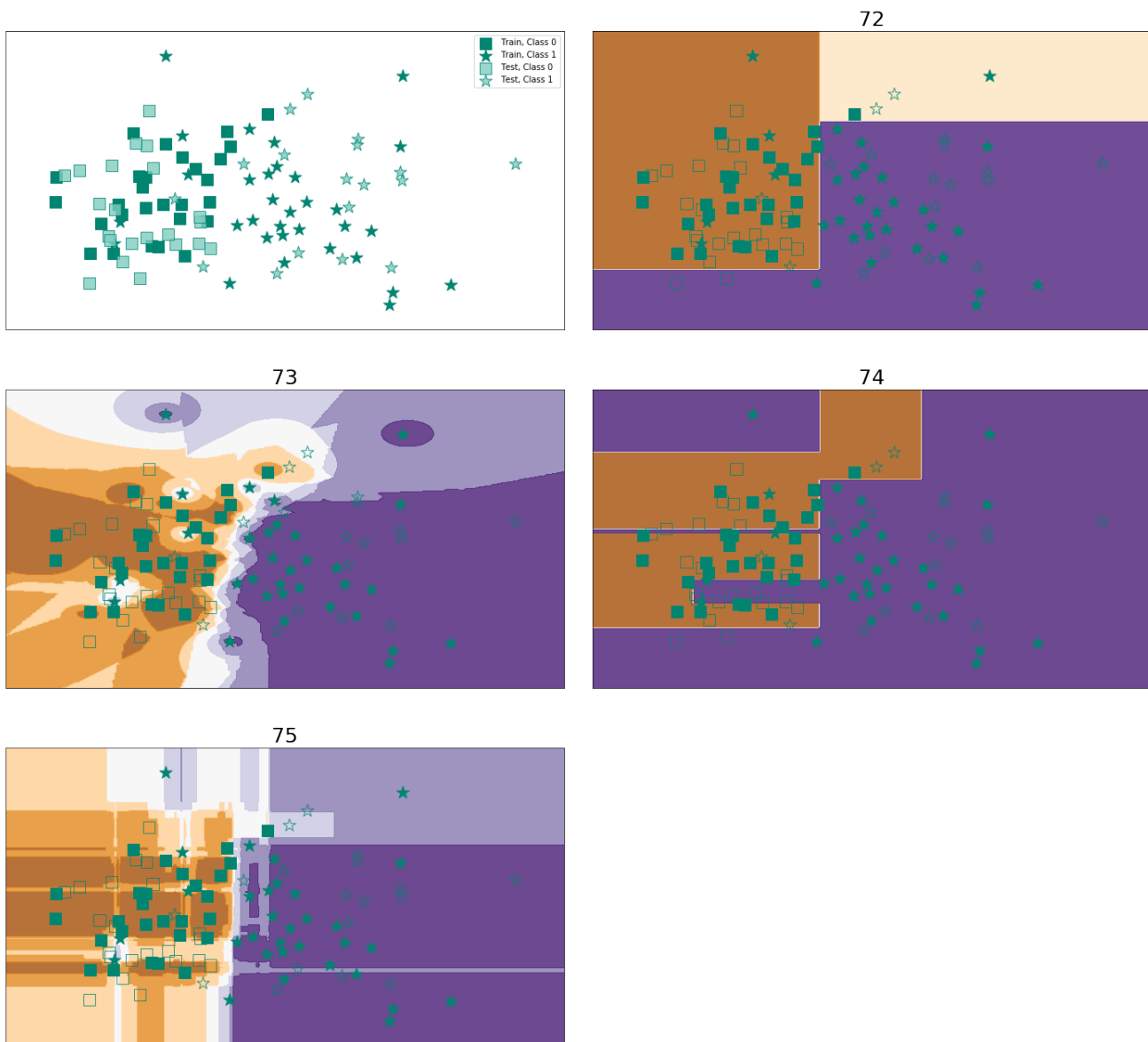
The first plot is the original dataset. The next four plots numbered as Questions¹ were generated by one of the algorithm configurations below. Fill in the appropriate box in the multiple choice bubble sheet at the end of the test booklet. (A) Naive Bayes (B) LDA (C) kNN: k=1 (D) kNN: k=5 (E) Decision Tree: maxDepth=6



¹Don't worry that the question numbers are higher, there's still room on the bubble sheet. Empty question numbers won't be used.

The first plot is the original dataset. The next four plots numbered as Questions² were generated by one of the algorithm configurations below. Fill in the appropriate box in the multiple choice bubble sheet at the end of the test booklet. (A) RF: numTrees=100, maxDepth=6 (B) Decision Tree: max depth=2 (C) Decision Tree: maxDepth=6 (D) kNN: k=5 dist=Euc (E) LDA

Complex Classifiers on Random Binary Data(dataset 2)



²Don't worry that the question numbers are higher, there's still room on the bubble sheet. Empty question numbers won't be used.

Extra pages for answers or rough work. Please specify the question number here and the use of this page on the question page.

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